



### **Semester 2 Examinations 2013/ 2014**

**Exam Code(s)** 1MEC1, 1OA1, 3BS4, 3BMS2, 4BME1, 4BS3, 4FM2  
**Exam(s)** 3<sup>rd</sup> Science, 4<sup>th</sup> Science, 4<sup>th</sup> Financial Mathematics and Economics, 4<sup>th</sup> Mathematics and Education

**Module Code(s)** MP491  
**Module(s)** Non-linear Systems

**Paper No.** 1  
**Repeat Paper** No

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**Internal Examiner(s)** Dr P. T. Piironen  
Professor M. Destrade

**Instructions:** For full marks answer question 1 in Section A **and** TWO questions in Section B correctly.  
All questions carry the same weight.

***Duration*** 2hrs  
**No. of Pages** 5  
**Department(s)** Applied Mathematics  
**Course Co-ordinator(s)** Dr P. T. Piironen

**Requirements:**

Calculator Yes  
Statistical/ Log Tables Yes  
Release to Library Yes

**Section A** - Answer all questions (a)-(d). (5 Marks/question)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = 2x^2 - 8x + 6.$$

Find all equilibrium points and determine if they are stable or not.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point  $(x, y) = (0, 0)$  and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= y + x^2 - 1, \\ \frac{dy}{dt} &= y + x + 1. \end{aligned}$$

Find all equilibrium points and classify them.

- (d) Consider the non-linear map

$$x_{n+1} = (x_n)^2 - x_n - 2.$$

Find all fixed points and determine if they are stable or unstable.

**Section B** - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = (x - 1)(x^2 - 2x + \mu),$$

where  $\mu$  is a real parameter. Locate all equilibrium points, sketch the bifurcation diagram, and indicate the behaviour of the non-equilibrium solutions. Find all bifurcation points and classify them.

**10 Marks**

- (b) Consider the *Brusselator* given by the system of ordinary differential equations

$$\begin{aligned}\dot{x} &= a + x^2y - (b + 1)x, \\ \dot{y} &= bx - x^2y.\end{aligned}$$

Let  $a = 1$  and describe how the only equilibrium point varies with the parameter  $b$ . Find the bifurcation point  $(x, y, b)$  for which a Hopf bifurcation occurs. Describe how the nature of the equilibrium point changes at the Hopf bifurcation?

**10 Marks**

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= y^2 - x - 1, \\ \frac{dy}{dt} &= y^2 + x - 1.\end{aligned}$$

(a) Locate and classify all equilibrium points in the associated phase plane.

**4 Marks**

(b) Identify, and sketch separately, the isoclines corresponding to  $k = -1, 0, 1, \infty$ , where

$$\frac{dy}{dx} = k.$$

**12 Marks**

(c) By using all four isoclines in part (b), sketch a sufficient number of phase plane curves to illustrate the phase-plane portrait for this system.

**4 Marks**

4. Consider the Ricker Model, that describes the evolution of a population  $P$ , given by

$$P_{n+1} = R_0 P_n e^{-P_n},$$

where  $P_n \geq 0$  is the population at year  $n$  and  $R_0 > 0$  is the max reproduction rate.

- (a) Find all fixed points and investigate for what values of  $R_0$  the Ricker model has **one** and **two** fixed points, respectively.

**5 Marks**

- (b) Calculate the stabilities of the fixed points found in (a) when varying  $R_0$ .

**5 Marks**

- (c) Find the value of  $R_0$  for which the positive fixed point undergoes a *period-doubling bifurcation*.

**2 Marks**

- (d) Derive an expression that can be used to locate period-2 points for the Ricker model. Use the result in (b) to describe how you calculate the stability of a period-2 orbit in the Ricker model.

**NOTE: There is no need to explicitly calculate the period-2 points.**

**5 Marks**

- (e) If  $R_0$  is increased further, passed the point of the period-doubling bifurcation found in (c), the Ricker model will undergo a *period-doubling sequence to chaos*. Describe what is meant a by period-doubling sequence to chaos and sketch a bifurcation diagram showing this.

**3 Marks**